

LINE DESIGNATION

TO COME

LINE SERVICE CODE DESCRIPTION

TO COME

SPECIFICATION/MATERIAL INDEX

SSL316L STAINLESS STEEL
PVCPLASTIC (PVC)

VALVES

NORMALLY OPEN

VALVE TYPE/CODE

BALL/BA
GLOBE/GL
GATE/GA
NEEDLE/ND
CHECK/CK
THREE WAY/TW
RELIEF VALVE

NORMALLY CLOSED

N/A
N/A
N/A
N/A

PIPE LINE LEGEND

PRIMARY PROCESS LINE
SECONDARY PROCESS
FLEXIBLE HOSE
ELECTRICAL/INSTRUMENTATION SIGNAL

INSULATION SERVICE DESCRIPTION

HHOT
NNONE

C COLD

EQUIPMENT ABBREVIATIONS

AAGITATOR AND OTHER MIXING EQUIPMENTS
BBURNERS, COMBUSTION CHAMBER, THERMAL OXIDIZER
CCONVEYING EQUIPMENT & BULK BAG MATERIAL HANDLING EQUIPMENTS
DDRYERS
EGENERATOR
FFAN, COMPRESSOR, BLOWER
GGAS / SOLID SEPARATION EQUIPMENT
HHEAT EXCHANGER, HEATING COIL, HEATER, EVAPORATOR, CONDENSER, COOLER
JSTEAM GENERATOR, STEAM BOILER
KKILN & OTHER FURNACE
LLIQUID/SOLID SEPARATION EQUIPMENT
MSTATIC MIXER
MOTMOTOR
NNOISE REDUCTION EQUIPMENT
PPUMP
QAIR CONDITIONER & BUILDING VENT. SYSTEM
ROREVERSE OSMOSIS EQUIPMENT
SSILO, BIN (FOR SOLID)
TTANK, DRUM, PRESSURE VESSEL (LIQUID AND GAS)
TRAPSTEAM TRAP
ULIFTING EQUIPMENT
VVEHICLE
WWEIGHING EQUIPMENT
XFLARE, VENT, STACK, CHIMNEY
YLABORATORY & WORKSHOP EQUIPMENT, TOOLS FOR MAINTENANCE
ZPACKAGE, SKID OR EQUIPMENT NOT DEFINED BY ANOTHER LETTER

INSTRUMENTATION

MASS FLOW METER
ROTAMETER
INTEGRATED THERMAL MASS FLOW CONTROLLER
PADDLE/TURBINE FLOW METER
MAGNETIC FLOW METER

ACTUATORS

SOLENOID
DIAPHRAGM
REGULATOR
I TO P CONVERTER

CONTROL SYMBOLS AND ABBREVIATIONS

LOCALLY MOUNTED
PANEL MOUNTED

PROCESS CONTROL SYSTEM (PCS)
"X" SIGNIFIES ALARM FUNCTION
INTERLOCK VIA PCS
(SEE CAUSE & EFFECT)

INSTRUMENTATION ABBREVIATIONS

NONORMALLY OPEN
NENORMALLY CLOSED
FOFAIL OPEN
FCFAIL CLOSED
FLFAIL IN LAST POSITION
FIFAIL INDETERMINATE
LOLOCK OPEN
LCLOCK CLOSED
SPSTOP
STSTART

CLCLOSED
OPOPEN
TDTIME DELAY
STPSET POINT
RATIORATIO CONTROL
RSPREMOTE SET POINT
HHIGH ALARM/INTERLOCK
HHHIGH HIGH ALARM/INTERLOCK
LLOW ALARM/INTERLOCK
LLLOW LOW ALARM/INTERLOCK

OTHER SYMBOLS

REDUCER
ISOLATOR
FLANGE CONNECTION
UNION CONNECTION
RUPTURE DISC
MATERIAL OR CLASS CHANGE
LINE NUMBER CHANGE
IN-LINE STRAINER
Y-STRAINER
GENERIC FILTER
THREADED PLUG
BLIND FLANGE
RESTRICTION ORIFICE
CENTRIFUGAL PUMP/BLOWER
POSITIVE DISPLACEMENT PUMP
GENERIC HEAT EXCHANGER
BATTERY LIMIT
MODULE LIMIT
ELECTRIC MOTOR
HOSE CONNECTION
LIQUID TRAP
MAGNETIC TRAP

INSTRUMENT TAG NUMBER SYSTEM

EACH INSTRUMENT IS IDENTIFIED BY A COMBINATION OF LETTERS AND NUMBERS. THE LETTERS ARE SELECTED FROM TABLE-A TO CLASSIFY THE INSTRUMENT FUNCTION. THE LETTERS ARE FOLLOWED BY A LOOP NUMBER. THE FIRST DIGITS OF THE LOOP NUMBER ARE SELECTED FROM TABLE-B TO DESIGNATE THE PROCESS AREA OF THE INSTRUMENT. THE LAST DIGITS OF THE NUMBER WILL NUMERICALLY IDENTIFY IT.

THE AREA CODE AND LOOP IDENTIFICATION NUMBERS WILL BE COMMON TO ALL INSTRUMENTS WITHIN THAT LOOP. A LETTER SUFFIX IS ADDED TO DIFFERENTIATE BETWEEN INSTRUMENTS BEARING THE SAME LETTER PREFIX WITHIN ANY GIVEN LOOP.

EXAMPLE

L FIRST LETTER	IC SUCCEEDING LETTERS	0000 NUMBER	A SUFFIX IF REQ'D
FUNCTIONAL IDENTIFICATION		LOOP NUMBER	
INSTRUMENT TAG NUMBER			

TABLE A

LETTER IDENTIFICATION					
FIRST LETTER		SUCCEEDING LETTERS			
	MEASURED OR INITIATING VARIABLE	MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A	ANALYSIS				
B	BURNER FLAME				
C				CONTROL	
D	DENSITY OR SPECIFIC GRAVITY	DIFFERENTIAL			
E	VOLTAGE		PRIMARY ELEMENT		
F	FLOW	RATIO			
G	GAUGING				
H	HAND OPERATED				HIGH
I	CURRENT		INDICATE		
J	POWER	SCAN			
K	TIME			CONTROL STATION	
L	LEVEL		LIGHT (PILOT)		LOW
M	MOISTURE				MIDDLE OR INTERMEDIATE
N					
O	TORQUE		ORIFICE (RESTRICT)		
P	PRESSURE OR VACUUM		POINT (TEST CONNECTION)		
Q	QUANTITY	INTEGRATE OR TOTALIZE			
R	RADIOACTIVITY		RECORD OR PRINT		
S	SPEED OR FREQUENCY	SAFETY		SWITCH	
T	TEMPERATURE			TRANSMIT	
U	MULTIVARIABLE		MULTIFUNCTION		
V	VISCOSITY OR VIBRATION			VALVE OR DAMPER	
W	WEIGHT OR FORCE		WELL		
X	SEQUENCE		UNCLASSIFIED		
Y				RELAY OR COMPUTE	
Z	POSITION			DRIVE, ACTUATE OR UNCLASSIFIED FINAL CONTROL ELEMENT	

REFERENCE DRAWINGS

DRAWING NO.	DRAWING TITLE

NOTES:
1.

REV

REV	REVISION DESCRIPTION	DATE
A	PRELIMINARY – FOR CLIENT REVIEW	2024.06.10

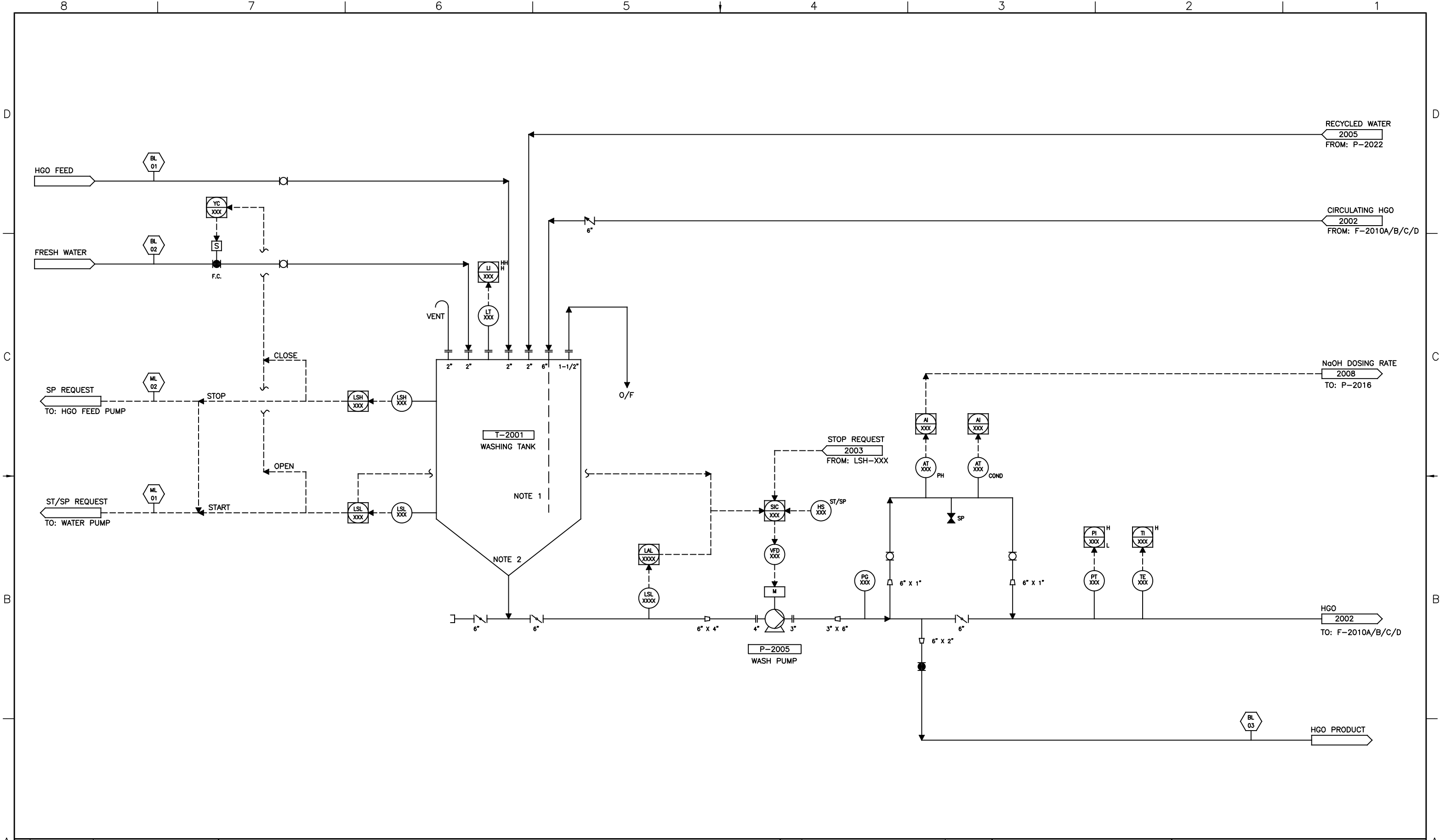
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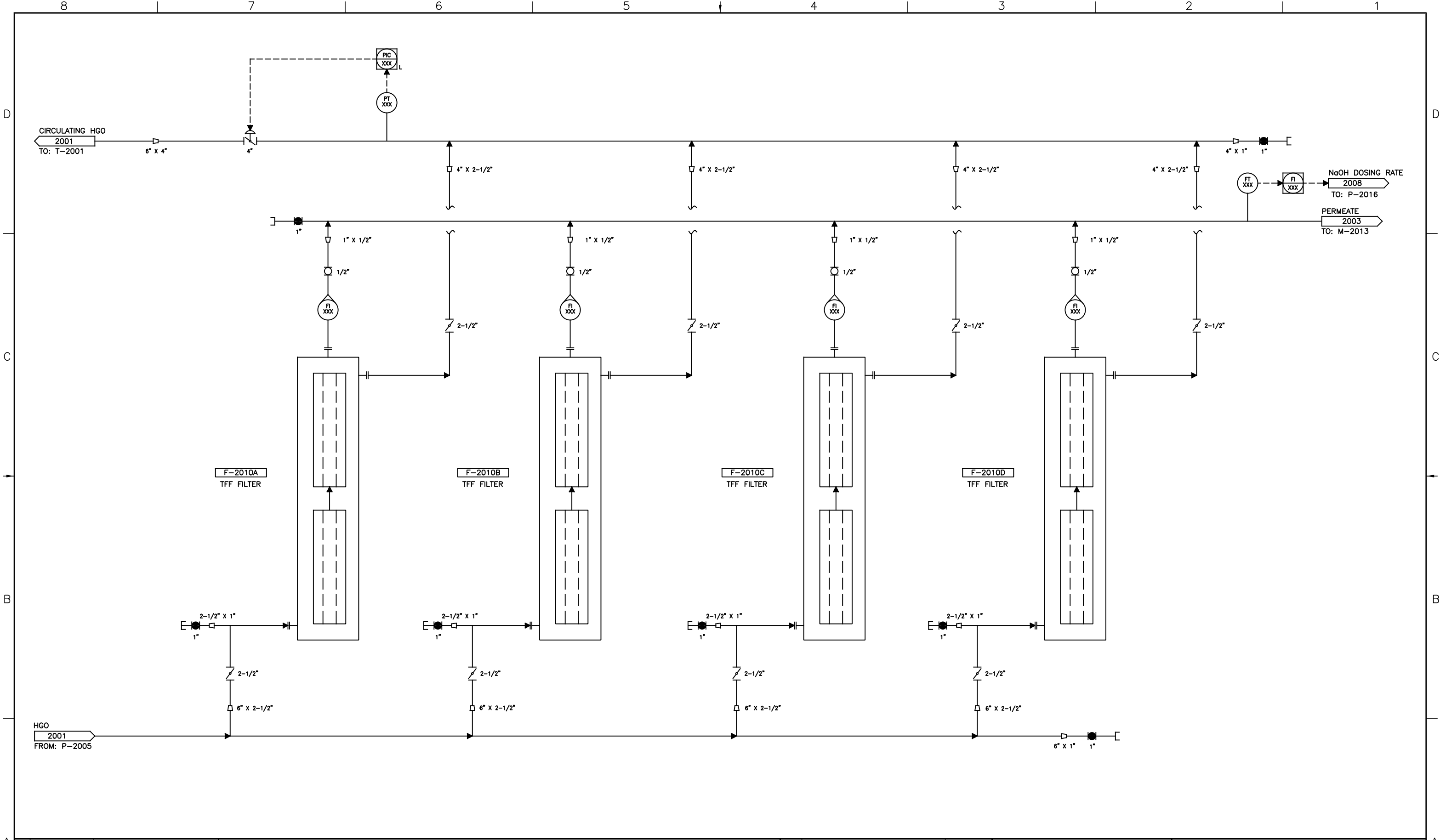
DESIGNED TL MAY 2024
DRAWN TL JUNE 2024
CHECKED AG JUNE 2024
APPROVED AG JUNE 2024

LEGEND SHEET
PIPING & INSTRUMENTATION DIAGRAM
SIZE DWG NO. B 1054ENG1-DWG-2000A
PRELIMINARY SCALE NTS

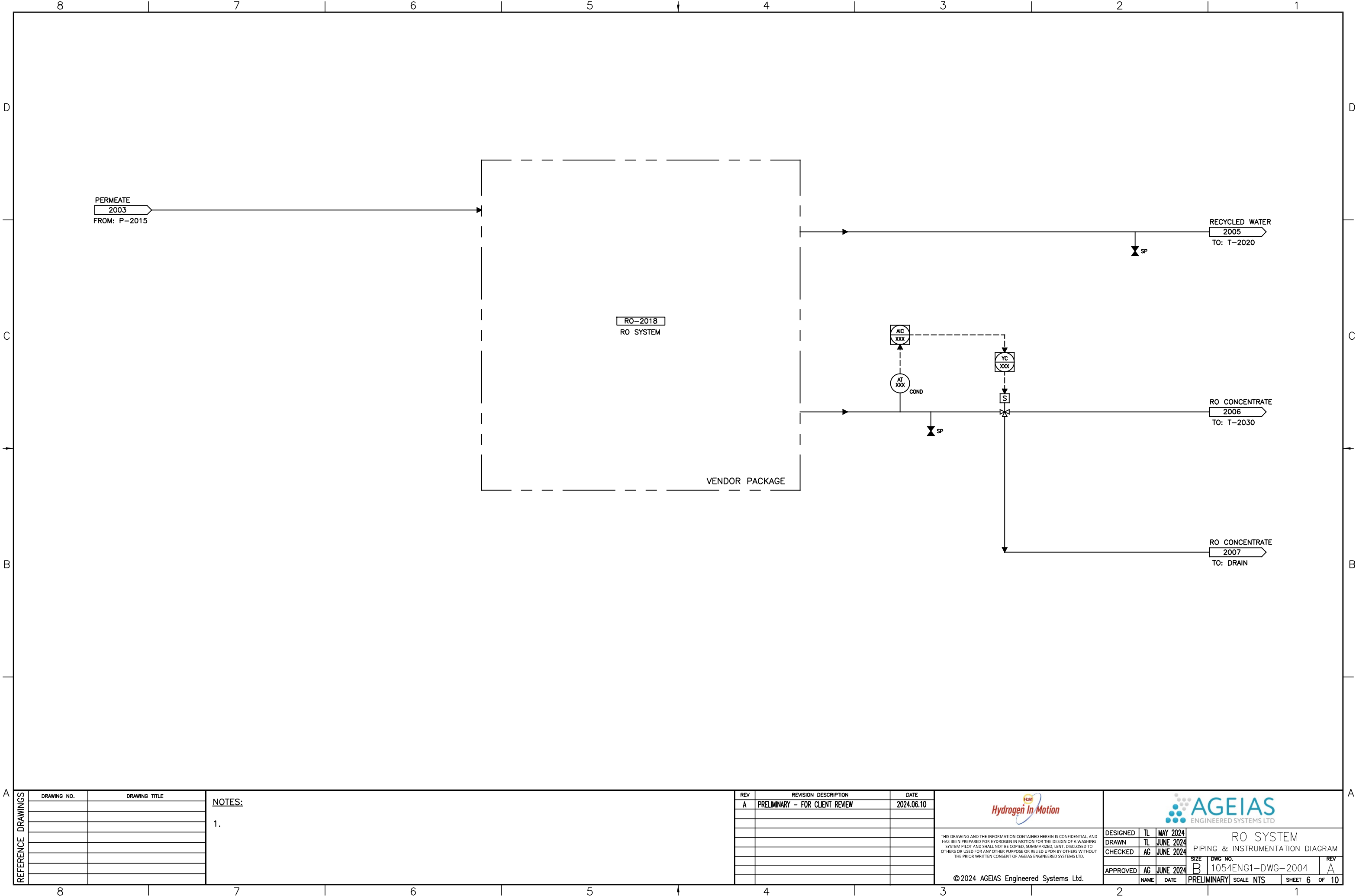
REV A
SHEET 1 OF 10



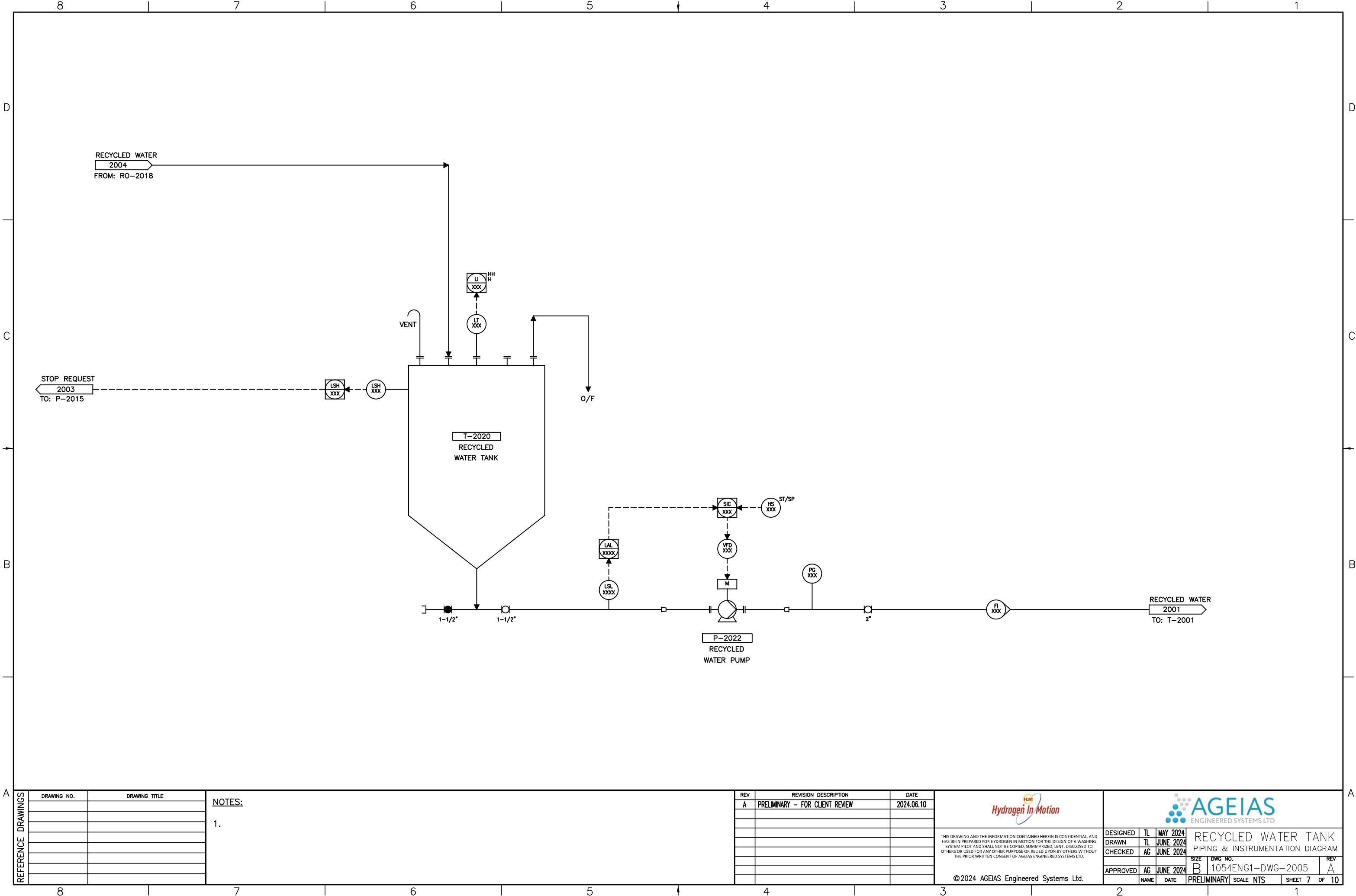
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								CHECKED	AG	JUNE 2024	SIZE	DWG NO.	REV
								APPROVED	AG	JUNE 2024	B	1054ENG1–DWG–2004	A
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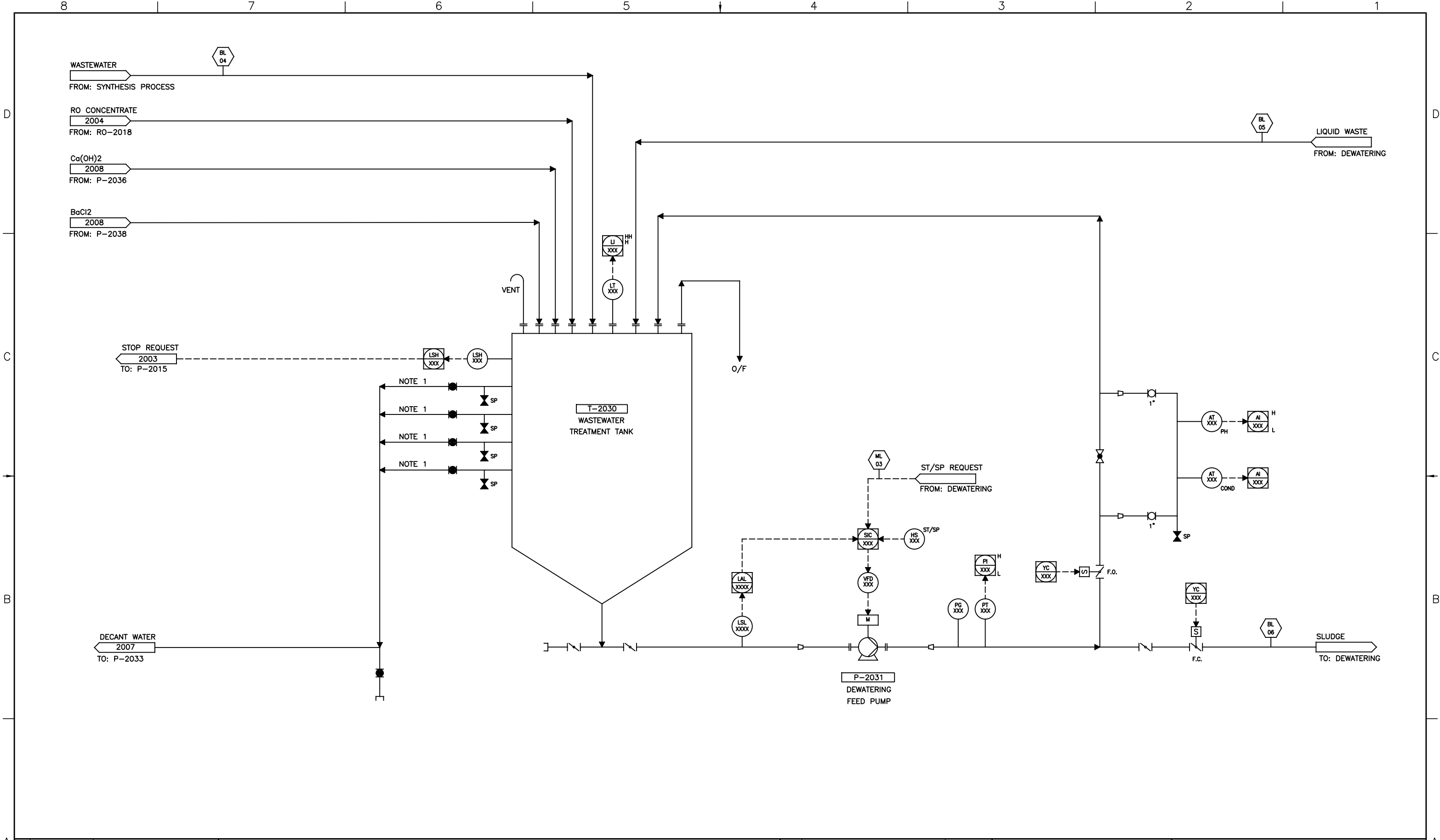
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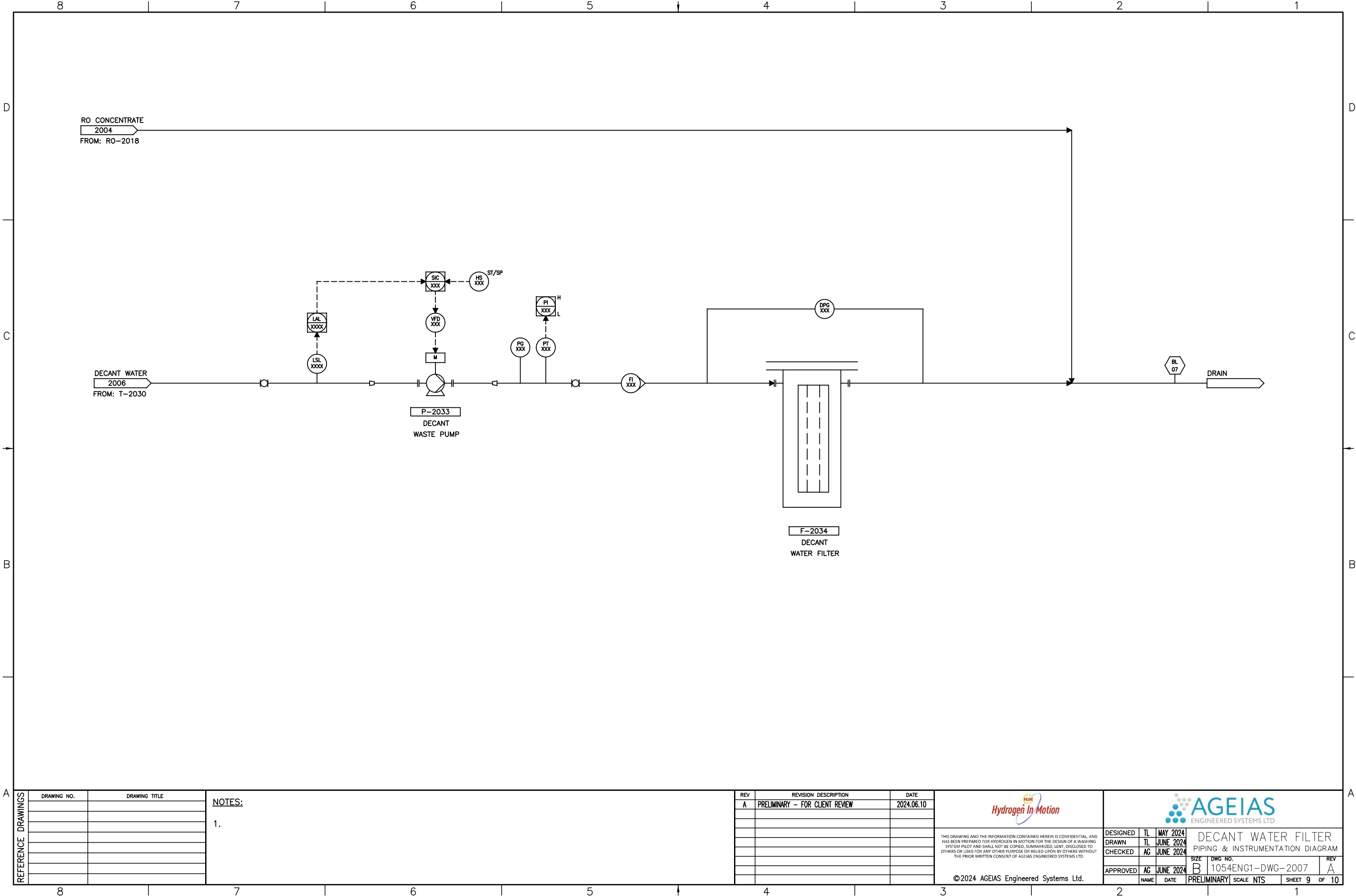
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				RECYCLED WATER TANK			
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DRAWN	TL	JUNE 2024					
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APPROVED	AG	JUNE 2024					
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